

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			2	1			1		
	(b)	(i)		Slows down the neutrons (1) which enables the fission reaction to happen (1)	2			2		
		(ii)		Fission reaction [in fast breeder] happens with fast neutrons / no need to slow neutrons down for fission reaction [so moderator not needed]			1	1		
	(c)			Indicative content: Advantages 1. Underground storage ensures a large distance between HLW and people. 2. Ionising radiation absorbed by the Earth so reduces exposure to people living on the surface. 3. Glass / containers / concrete are stable for thousands of years which reduces the risk of HLW leaking. 4. Employment for scientists/security workers etc who work at the HLW storage facility. 5. Easy to monitor the HLW during long term storage. Disadvantages 1. Need stable geological area to ensure HLW storage is safe. 2. Costly to construct, monitor and ensure high security of deep underground HLW storage area. 3. HLW has long half-life so storage facility needs to operate for 1000's of years. 4. Safety and security risks associated with transporting HLW to the storage facility. 5. NIMBY concerns.	6			6		

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				5–6 marks Detailed description of at least 3 advantages and 3 disadvantages. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3–4 marks A brief description of 2 advantages and 2 disadvantages. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1–2 marks A very brief description of either 1 or 2 advantages or 1 or 2 disadvantages or 1 advantage and 1 disadvantage. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 7 total	9	0	1	10	0	0